$\boldsymbol{f}(z) = \boldsymbol{S}z$) be zero. Applying Euler's method to the same system results in

$$z_{n+1} = z_n + h\boldsymbol{S}z_n = (\boldsymbol{I} + h\boldsymbol{S})z_n,$$

and the condition for Euler's method to conserve volume is that $\det(\boldsymbol{I} + h\boldsymbol{S}) = 1$. The conditions for volume preservation by the flow map and its Euler approximation are essentially unrelated. Thus Euler's method does not in general conserve phase space volume (it conserves volume only in very special cases—see Exercise 6).

Yet it turns out that there are some numerical methods that always conserve the volume when this is conserved by the flow of the differential equations, or for some particular classes of differential equations. For example, consider the planar system

$$\begin{aligned} \dot{u} &= f(u, v), \\ \dot{v} &= g(u, v), \end{aligned}$$

satisfying the condition $\frac{\partial f}{\partial u} + \frac{\partial g}{\partial v} = 0$, and the asymmetrical variant of Euler's method defined by

$$\begin{aligned} u_{n+1} &= u_n + hf(u_{n+1}, v_n), \\ v_{n+1} &= v_n + hg(u_{n+1}, v_n). \end{aligned}$$

The components of the Jacobian matrix can be obtained using implicit differentiation, they are:

$$a_{11} = \frac{\partial u_{n+1}}{\partial u_n}, \quad a_{12} = \frac{\partial u_{n+1}}{\partial v_n}, \quad a_{21} = \frac{\partial v_{n+1}}{\partial u_n}, \quad a_{22} = \frac{\partial v_{n+1}}{\partial v_n}.$$

So

$$a_{11} = 1 + hf_u a_{11}, \quad a_{12} = hf_u a_{12} + hf_v, \quad a_{21} = hg_u a_{11}, \quad a_{22} = 1 + hg_u a_{12} + hg_v.$$

Solving for the various entries we have

$$\mathscr{G}_h' = \begin{bmatrix} 1/(1 - hf_u) & hf_v/(1 - hf_u) \\ hg_u/(1 - hf_u) & 1 + hg_v + h^2 g_u f_v/(1 - hf_u) \end{bmatrix},$$

and calculating the determinant of the Jacobian results in

$$\det \mathscr{G}_h' = \frac{1 + hg_v}{1 - hf_u}.$$

In the event that the vector field is divergence free, we have $f_u + g_v = 0$ which implies that the numerator and denominator are identical, and it follows that